

MANIPULATIONS/TASKS PROTOCOL DOCUMENT

Listening Comprehension Study

1. Research Design

This study uses a 4×4 within-subjects Latin square design. The two factors are:

- **Topic** (4 levels): birds, fruits, stars, musical instruments
- **Text quality** (4 levels): control, repeat_short, repeat_long, distractor

Each participant reads 4 passages — one per topic, one per text quality condition — such that no topic or text quality condition is repeated across the session. Assignment of text quality conditions to topics is determined by Latin square randomization, counterbalanced across participants. Passage presentation order is randomized per participant.

Control	Baseline passage; key information presented once at standard length (~1:30)
Repeat_short	Same length as control; key information repeated within the passage
Distractor	Longer than control; contains additional non-key information (~4:00)
Repeat_long	Same length as distractor; key information repeated within the passage

Because each participant reads exactly one passage per text quality condition, total reading time is consistent across participants regardless of condition assignment.

Participants will be recruited through Prolific and complete the study remotely via Qualtrics. The study consists of a single session.

Sample sizes per cell are to be determined by power analysis on pilot study prior to data collection. We will recruit a minimum of 50 and a maximum of 500.

2. Description of Task

Participants complete the following steps in a single session:

1. Read a consent form and provide consent
2. For each of 4 passages (order randomized):
 - Listen to the passage once; passage will start playing when participants click “Start audio” and will play once
 - Answer multiple-choice comprehension questions about facts in the passage
 - Answer questions about their experience reading the passage
3. Receive a debriefing form

This study uses a standard listening comprehension paradigm, which is not expected to present any risks to participants beyond those encountered in everyday life.

3. Associated Materials

The following materials are attached:

- Passage stimuli (all 16 passages: 4 topics × 4 text quality conditions)
- Qualtrics survey (participant-facing instructions, passage presentation, comprehension questions, experience questions, debriefing form)